

**WEEK 2 – Supervised Learning: Regression & Classification**

SL No	Week	Session-1 (3 Hours)	Detailed Subtopics	Session-2 (3 Hours)	Detailed Subtopics
2	Week 2	Regression & Classification Models	Simple and multiple linear regression, Cost function (MSE), Gradient descent optimization, Logistic regression, Bias-variance trade-off	Model Training & Validation	Train-test split, Cross-validation techniques, Feature scaling, Model comparison strategies, Interpretation of results

CONTENTS (Week 2 – Session1)

1. Introduction to Supervised Learning
2. Regression vs Classification
3. Linear Regression
 - Simple Linear Regression
 - Multiple Linear Regression
4. Cost Function – Mean Squared Error (MSE)
5. Gradient Descent Optimization
6. Logistic Regression
7. Bias–Variance Trade-Off
8. Summary
9. Q & A

1. Introduction to Supervised Learning

Supervised Learning is a type of Machine Learning where the model is trained using **labeled data**.

Each training example consists of:

- **Input features (X)**
- **Target output (Y)**

The objective is to learn a mapping function:

$$f(X) \rightarrow Y$$

so that the model can accurately predict outputs for unseen data.

Key Characteristics

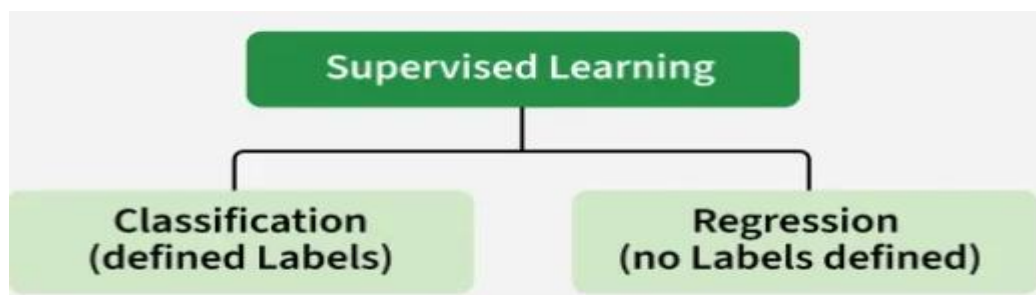
- Requires labeled datasets
- Learning is guided by known outputs
- Used for Regression and classification problems

Real-World Examples

- House price prediction
- Disease diagnosis
- Student performance prediction

2 Regression vs Classification

Supervised learning problems are broadly divided into **Regression** and **Classification**.



User ID	Gender	Age	Salary	Purchased	Temperature	Pressure	Relative Humidity	Wind Direction	Wind Speed
15624510	Male	19	19000	0	10.69261758	986.882019	54.19337313	195.7150879	3.278597116
15810944	Male	35	20000	1	13.59184184	987.8729248	48.0648859	189.2951202	2.909167767
15668575	Female	26	43000	0	17.70494885	988.1119385	39.11965597	192.9273834	2.973036289
15603246	Female	27	57000	0	20.95430404	987.8500366	30.66273218	202.0752869	2.965289593
15804002	Male	19	76000	1	22.9278274	987.2833862	26.06723423	210.6589203	2.798230886
15728773	Male	27	58000	1	24.04233986	986.2907104	23.46918024	221.1188507	2.627005816
15598044	Female	27	84000	0	24.41475295	985.2338867	22.25082295	233.7911987	2.448749781
15694829	Female	32	150000	1	23.93361956	984.8914795	22.35178837	244.3504333	2.454271793
15600575	Male	25	33000	1	22.68800023	984.8461304	23.7538641	253.0864716	2.418341875
15727311	Female	35	65000	0	20.56425726	984.8380737	27.07867944	264.5071106	2.318677425
15570769	Female	26	80000	1	17.76400389	985.4262085	33.54900114	280.7827454	2.343950987
15606274	Female	26	52000	0	11.25680746	988.9386597	53.74139903	68.15406036	1.650191426
15746139	Male	20	86000	1	14.37810685	989.6819458	40.70884681	72.62069702	1.553469896
15704987	Male	32	18000	0	18.45114201	990.2960205	30.85038484	71.70604706	1.005017161
15628972	Male	18	82000	0	22.54895853	989.9562988	22.81738811	44.66042709	0.264133632
15697686	Male	29	80000	0	24.23155922	988.796875	19.74790765	318.3214111	0.329656571
15733883	Male	47	25000	1					

Figure A: CLASSIFICATION

Figure B: REGRESSION

Aspect	Regression	Classification
Output Type	Continuous	Discrete
Example Output	Price, temperature	Yes/No, Class labels
Algorithms	Linear Regression	Logistic Regression, k-NN
Evaluation	MSE, RMSE	Accuracy, Precision, Recall, Confusion matrix

3. Linear Regression

Linear Regression models the relationship between input variables and a continuous output using a straight line.

3.1 Simple Linear Regression

Simple Linear Regression uses **one independent variable**.

Mathematical Model

$$y = mx + c$$

Where:

- $y \rightarrow$ predicted output
- $x \rightarrow$ input feature
- $m \rightarrow$ slope
- $c \rightarrow$ intercept

Example

Predicting salary based on years of experience.

Introduction to Linear Regression



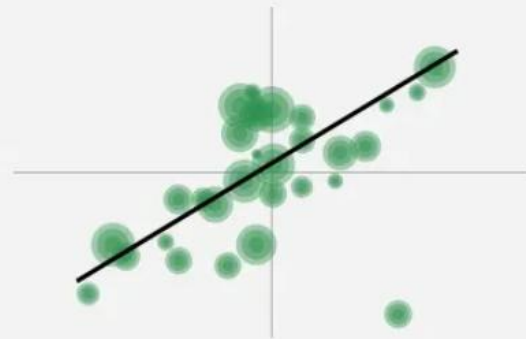
Simple model for predicting continuous values.



It finds the relationship between input features and output.



It is used in forecasting and predicting salaries based on experience.



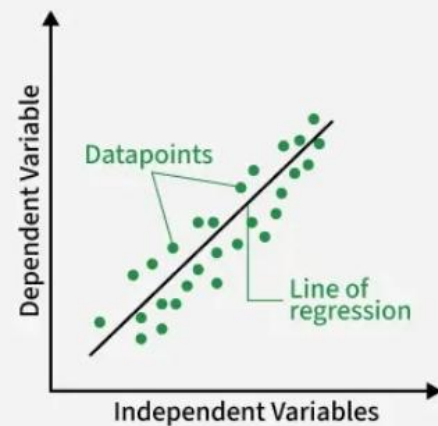
How Does Linear Regression Work?



Finds the best-fitting line by minimizing prediction errors (least squares method)



It calculates coefficients for variables that minimize the error in predictions.



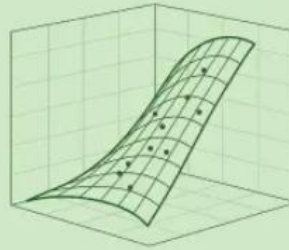
Types of Linear Regression

Simple Linear Regression



Predicts the dependent variable using a single independent variable.

Multiple Linear Regression



Uses two or more independent variables to predict the dependent variable.

Real-World Use Cases of Linear Regression



Stock Market Prediction



Real Estate Price Prediction



Medical Risk Prediction



Sales Forecasting

Challenges in Linear Regression



Overfitting



Multicollinearity



Assumptions of Linearity



Sensitive to Outliers

Best Fit Line in Linear Regression

In linear regression, the best-fit line is the straight line that most accurately represents the relationship between the independent variable (input) and the dependent variable (output). It is the line that minimizes the difference between the actual data points and the predicted values from the model.

Hypothesis function in Linear Regression

In linear regression, the hypothesis function is the equation used to make predictions about the dependent variable based on the independent variables. It represents the relationship between the input features and the target output.

For a simple case with one independent variable, the hypothesis function is:

$$h(x) = \beta_0 + \beta_1 x$$

Where:

- $h(x)$ or $(\hat{y})$ is the predicted value of the dependent variable (y).
- x is the independent variable.
- β_0 is the intercept, representing the value of y when x is 0.
- β_1 is the slope, indicating how much y changes for each unit change in x .

3.2 Multiple Linear Regression

Multiple Linear Regression uses **more than one independent variable**.

Mathematical Model

$$y = w_1 x_1 + w_2 x_2 + \dots + w_n x_n + b$$

Where:

- $x_1, x_2, \dots \rightarrow$ input features
- $w_1, w_2, \dots \rightarrow$ weights
- $b \rightarrow$ bias

Example

Predicting house price using:

- Area
- Number of bedrooms
- Location

For multiple linear regression (with more than one independent variable), the hypothesis function expands to:

$$h(x_1, x_2, \dots, x_k) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_k x_k$$

Where:

- $x_1, x_2, \dots, x_k$ are the independent variables.
- β_0 is the intercept.
- $\beta_1, \beta_2, \dots, \beta_k$ are the coefficients, representing the influence of each respective independent variable on the predicted output.

3.3 Cost Function

Cost function for Linear Regression

In Linear Regression, the cost function measures how far the predicted values ($\hat{Y}$) are from the actual values (Y). It helps identify and reduce errors to find the best-fit line. The most common cost function used is Mean Squared Error (MSE), which calculates the average of squared differences between actual and predicted values:

$$\text{Cost function}(J) = \frac{1}{n} \sum_n^i (\hat{y}_i - y_i)^2$$

Where:

- $y_i \rightarrow$ actual value
- $\hat{y}_i \rightarrow$ predicted value
- $n \rightarrow$ number of samples

Why Squared Error?

- Penalizes large errors more
- Differentiable → suitable for optimization

3.4 Gradient Descent for Linear Regression

Gradient descent is an optimization technique used to train a linear regression model by minimizing the prediction error. It works by starting with random model parameters and repeatedly adjusting them to reduce the difference between predicted and actual values.

Core Idea

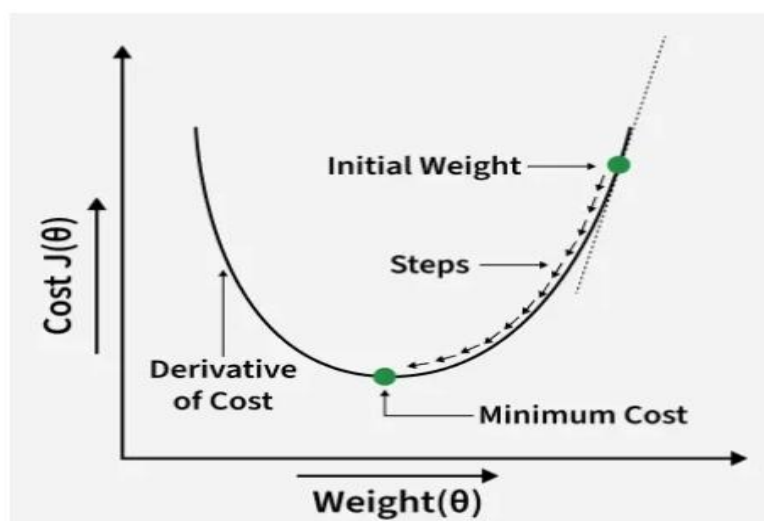
- Compute gradient of the loss function
- Move parameters in the **opposite direction of the gradient**

Update Rule

$$w = w - \alpha \frac{\partial J(w)}{\partial w}$$

Where:

- $w \rightarrow$ model parameters
- $\alpha \rightarrow$ learning rate
- $J(w) \rightarrow$ cost function



How it works:

- Start with random values for slope and intercept.
- Calculate the error between predicted and actual values.
- Find how much each parameter contributes to the error (gradient).
- Update the parameters in the direction that reduces the error.
- Repeat until the error is as small as possible.

This helps the model find the best-fit line for the data.

Types of Gradient Descent

- Batch Gradient Descent
- Stochastic Gradient Descent
- Mini-Batch Gradient Descent

3.5 Evaluation Metrics for Linear Regression

A variety of evaluation measures can be used to determine the strength of any linear regression model. These assessment metrics often give an indication of how well the model is producing the observed outputs.

The most common measurements are:

a. Mean Square Error (MSE)

Mean Squared Error (MSE) is an evaluation metric that calculates the average of the squared differences between the actual and predicted values for all the data points. The difference is squared to ensure that negative and positive differences don't cancel each other out.

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Here,

- n is the number of data points.

- y_i is the actual or observed value for the i^{th} data point.
- $\hat{y}_i$ is the predicted value for the i^{th} data point.

MSE is a way to quantify the accuracy of a model's predictions. MSE is sensitive to outliers as large errors contribute significantly to the overall score.

b. Mean Absolute Error (MAE)

Mean Absolute Error is an evaluation metric used to calculate the accuracy of a regression model. MAE measures the average absolute difference between the predicted values and actual values.

Mathematically MAE is expressed as:

$$MAE = \frac{1}{n} \sum_{i=1}^n |Y_i - \hat{Y}_i|$$

Here,

- n is the number of observations
- Y_i represents the actual values.
- $\hat{Y}_i$ represents the predicted values

Lower MAE value indicates better model performance. It is not sensitive to the outliers as we consider absolute differences.

c. Root Mean Squared Error (RMSE)

The square root of the residuals' variance is the Root Mean Squared Error. It describes how well the observed data points match the expected values or the model's absolute fit to the data. In mathematical notation, it can be expressed as:

$$RMSE = \sqrt{\frac{RSS}{n}} = \sqrt{\frac{\sum_{i=1}^n (y_i^{actual} - y_i^{predicted})^2}{n}}$$

Where:

- n : Number of observations

- y_i : Actual value
- $\hat{y}_i$: Predicted value

RMSE is in the same unit as the target variable and highlights larger errors more clearly.

d. Coefficient of Determination (R-squared)

R-Squared is a statistic that indicates how much variation the developed model can explain or capture. It is always in the range of 0 to 1. In general, the better the model matches the data, the greater the R-squared number.

In mathematical notation, it can be expressed as:

$$R^2 = 1 - \left(\frac{RSS}{TSS} \right)$$

- **Residual sum of Squares (RSS):** The sum of squares of the residual for each data point in the plot or data is known as the residual sum of squares or RSS. It is a measurement of the difference between the output that was observed and what was anticipated.

$$RSS = \sum_{i=1}^n (y_i - b_0 - b_1 x_i)^2$$

- **Total Sum of Squares (TSS):** The sum of the data points' errors from the answer variable's mean is known as the total sum of squares or TSS.

$$TSS = \sum_{i=1}^n (y - \bar{y}_i)^2.$$

R squared metric is a measure of the proportion of variance in the dependent variable that is explained the independent variables in the model.

e. Adjusted R-Squared Error

Adjusted R^2 measures the proportion of variance in the dependent variable that is explained by independent variables in a regression model. Adjusted R-square accounts the number of predictors in the model and penalizes the model for including irrelevant

predictors that don't contribute significantly to explain the variance in the dependent variables.

Mathematically, adjusted R^2 is expressed as:

$$Adjusted R^2 = 1 - \left(\frac{(1 - R^2) \cdot (n - 1)}{n - k - 1} \right)$$

Here,

- n is the number of observations
- k is the number of predictors in the model
- R^2 is coefficient of determination

It penalizes the inclusion of unnecessary predictors, helping to prevent overfitting.

3.6 Regularization Techniques for Linear Models

1. Lasso Regression (L1 Regularization)

Lasso Regression is a technique used for regularizing a linear regression model, it adds a penalty term to the linear regression objective function to prevent overfitting.

The objective function after applying lasso regression is:

$$J(\theta) = \frac{1}{2m} \sum_{i=1}^m (\hat{y}_i - y_i)^2 + \lambda \sum_{j=1}^n |\theta_j|$$

- the first term is the least squares loss, representing the squared difference between predicted and actual values.
- the second term is the L1 regularization term, it penalizes the sum of absolute values of the regression coefficient θ_j .

2. Ridge Regression (L2 Regularization)

Ridge regression is a linear regression technique that adds a regularization term to the standard linear objective. Again, the goal is to prevent overfitting by penalizing large coefficient in linear regression equation. It useful when the dataset has multicollinearity where predictor variables are highly correlated.

The objective function after applying ridge regression is:

$$J(\theta) = \frac{1}{2m} \sum_{i=1}^m (\hat{y}_i - y_i)^2 + \lambda \sum_{j=1}^n \theta_j^2$$

3.7 Why Linear Regression is Important

Here's why linear regression is important:

- **Simplicity and Interpretability:** It's easy to understand and interpret, making it a starting point for learning about machine learning.
- **Predictive Ability:** Helps predict future outcomes based on past data, making it useful in various fields like finance, healthcare and marketing.
- **Basis for Other Models:** Many advanced algorithms, like logistic regression or neural networks, build on the concepts of linear regression.
- **Analysis:** It provides insights into relationships between variables.

Advantages

- Linear regression is a relatively simple algorithm, making it easy to understand and implement. The coefficients of the linear regression model can be interpreted as the change in the dependent variable for a one-unit change in the independent variable, providing insights into the relationships between variables.
- Linear regression is computationally efficient and can handle large datasets effectively. It can be trained quickly on large datasets, making it suitable for real-time applications.

Limitations

- Linear regression assumes a linear relationship between the dependent and independent variables. If the relationship is not linear, the model may not perform well.

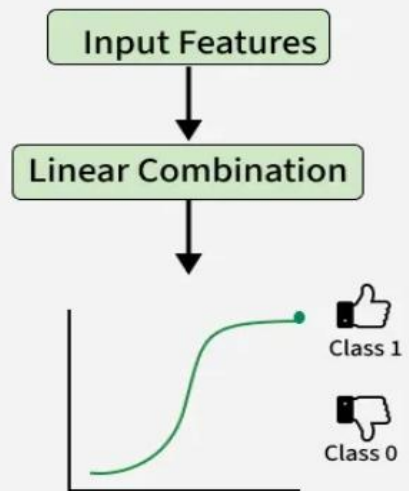
- Linear regression is sensitive to multicollinearity, which occurs when there is a high correlation between independent variables. Multicollinearity can inflate the variance of the coefficients and lead to unstable model predictions.

4. Logistic Regression

Logistic Regression is a **classification algorithm**, despite its name.

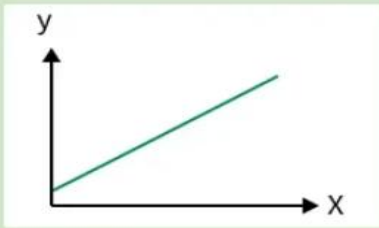
What is Logistic Regression?

- Predicts the probability of a binary outcome (Yes/No, 0/1)
- Uses the sigmoid function to map inputs to probabilities (0 to 1)
- Ideal for classification tasks



Linear Regression

- Predicts continuous values
- Uses best-fit line
- Solves regression problems



vs

Logistic Regression

- Predicts categorical classes
- Uses sigmoid S-curve
- Solves classification problems



Types of Logistic Regression

Binomial

Two classes (0 or 1)



Multinomial

More than two unordered classes (cat, dog, sheep)



Ordinal

More than two ordered classes (low, medium, high)



Output

- Produces probabilities between **0 and 1**

Sigmoid Function

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

1. The sigmoid function is an important part of logistic regression which is used to convert the raw output of the model into a probability value between 0 and 1.
2. This function takes any real number and maps it into the range 0 to 1 forming an "S" shaped curve called the sigmoid curve or logistic curve. Because probabilities must lie between 0 and 1, the sigmoid function is perfect for this purpose.

3. In logistic regression, we use a threshold value usually 0.5 to decide the class label.

- If the sigmoid output is same or above the threshold, the input is classified as Class 1.
- If it is below the threshold, the input is classified as Class 0.

This approach helps to transform continuous input values into meaningful class predictions.

How does Logistic Regression work?

Logistic regression model transforms the linear regression function continuous value output into categorical value output using a sigmoid function which maps any real-valued set of independent variables input into a value between 0 and 1. This function is known as the logistic function.

Suppose we have input features represented as a matrix:

$$X = \begin{bmatrix} x_{11} & \dots & x_{1m} \\ x_{21} & \dots & x_{2m} \\ \vdots & \ddots & \vdots \\ x_{n1} & \dots & x_{nm} \end{bmatrix}$$

and the dependent variable is Y having only binary value i.e 0 or 1.

$$Y = \begin{cases} 0 & \text{if Class1} \\ 1 & \text{if Class2} \end{cases}$$

then, apply the multi-linear function to the input variables X .

$$z = \left(\sum_{i=1}^n w_i x_i \right) + b$$

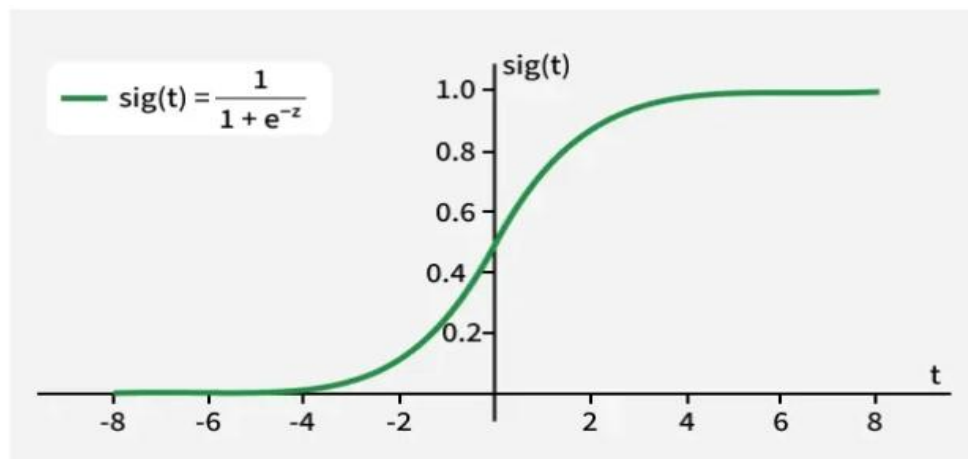
Here x_i is the i th observation of X , $w_i = [w_1, w_2, w_3, \dots, w_m]$ is the weights or Coefficient and b is the bias term also known as intercept. Simply this can be represented as the dot product of weight and bias.

$$z = w \cdot X + b$$

At this stage, z is a continuous value from the linear regression. Logistic regression then applies the sigmoid function to z to convert it into a probability between 0 and 1 which can be used to predict the class.

Now we use the sigmoid function where the input will be z and we find the probability between 0 and 1. i.e. predicted y .

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$



Sigmoid function

As shown above the sigmoid function converts the continuous variable data into the probability i.e between 0 and 1.

- $\sigma(z)$ tends towards 1 as $z \rightarrow \infty$
- $\sigma(z)$ tends towards 0 as $z \rightarrow -\infty$
- $\sigma(z)$ is always bounded between 0 and 1

where the probability of being a class can be measured as:

$$P(y = 1) = \sigma(z)$$

$$P(y = 0) = 1 - \sigma(z)$$

Applications

- Spam detection
- Disease classification

Terms in Logistic Regression

Here are some common terms involved in logistic regression:

1. **Independent Variables:** These are the input features or predictor variables used to make predictions about the dependent variable.
2. **Dependent Variable:** This is the target variable that we aim to predict. In logistic regression, the dependent variable is categorical.
3. **Logistic Function:** This function transforms the independent variables into a probability between 0 and 1 which represents the likelihood that the dependent variable is either 0 or 1.
4. **Odds:** This is the ratio of the probability of an event happening to the probability of it not happening. It differs from probability because probability is the ratio of occurrences to total possibilities.
5. **Log-Odds (Logit):** The natural logarithm of the odds. In logistic regression, the log-odds are modeled as a linear combination of the independent variables and the intercept.
6. **Coefficient:** These are the parameters estimated by the logistic regression model which shows how strongly the independent variables affect the dependent variable.
7. **Intercept:** The constant term in the logistic regression model which represents the log-odds when all independent variables are equal to zero.
8. **Maximum Likelihood Estimation (MLE):** This method is used to estimate the coefficients of the logistic regression model by maximizing the likelihood of observing the given data.

Differences Between Linear and Logistic Regression

Logistic regression and linear regression differ in their application and output.

Here's a comparison:

Aspect	Linear Regression	Logistic Regression
Definition	Linear regression is used to predict the continuous dependent variable using a given set of independent variables.	Logistic regression is used to predict the categorical dependent variable using a given set of independent variables.
Problem Type	It is used for solving regression problem.	It is used for solving classification problems.
Output Type	In this we predict the value of continuous variables.	In this we predict values of categorical variables.
Curve/Model Fitting	In this we find best fit line.	In this we find S-Curve.
Estimation Method	Least square estimation method is used for estimation of accuracy.	Maximum likelihood estimation method is used for estimation of accuracy.
Output Example	The output must be continuous value such as price, age etc.	Output must be categorical value such as 0 or 1, Yes or No, etc.
Relationship Requirement	It required linear relationship between dependent and independent variables.	It not required linear relationship.
Collinearity	There may be collinearity between the independent variables.	There should be little to no collinearity between independent variables.

5) Bias–Variance Trade-Off

Bias is the error caused by **over-simplifying the model**.

Characteristics

- Model makes strong assumptions
- Fails to capture complex patterns
- High training error and high test error

Example

Using linear regression to fit a non-linear curve.

Result: Model predicts poorly everywhere.

Variance is the error caused by the model being **too sensitive to training data**.

Characteristics

- Model learns noise in data
- Very different models for small data changes
- Low training error, high test error

Example

A deep decision tree that perfectly fits training data but fails on new data.

Underfitting happens when the model is too simple to learn data patterns.

Characteristics

- **High bias**
- Poor performance on both training and test data

Example

Predicting house prices using only size, ignoring location and age.

Overfitting

Overfitting happens when the model learns training data too well, including noise.

Characteristics

- **High variance**
- Excellent training accuracy
- Poor test accuracy

Example

Memorizing exam answers instead of understanding concepts.

Bias–Variance Trade-off

Model Type	Bias	Variance
Simple model	High	Low
Complex model	Low	High

Summary

- Supervised learning uses labeled data
- Regression predicts continuous values
- Classification predicts discrete labels
- Linear regression models relationships
- MSE measures prediction error
- Gradient descent optimizes parameters
- Logistic regression handles classification
- Bias–variance trade-off affects generalization

Q&A

Q1) You are given past loan data with final approval decisions already present. Why is this not an unsupervised problem, and what does the model actually learn from such data?

Answer: This is supervised learning because correct outputs are known. The model learns a mapping between input features and known outcomes so it can predict decisions for new applicants.

Q2) A model predicts crop yield as 3.8 tons per acre. Later, another model predicts whether the crop will fail or succeed. Why are these two different learning problems?

Answer: The first is regression because it predicts a continuous numeric value. The second is classification because it predicts a discrete category.

Q3) A linear regression model fits a straight line but gives large errors for curved data patterns. What limitation of the model is exposed here?

Answer: Linear regression assumes a linear relationship between variables. When the true relationship is non-linear, the model shows high bias and underfits the data.

Q4) Two regression models produce predictions, but one has occasional very large errors. Which error measure exposes this clearly and why?

Answer: Mean Squared Error exposes this because it squares errors, giving more weight to large deviations.

Q5) Why are model parameters updated repeatedly instead of being set once during training?

Answer:

Gradient descent updates parameters iteratively so the model gradually moves toward minimizing prediction error rather than guessing optimal values in one step.

Q6) A model predicts whether an email is spam or not but outputs probabilities instead of fixed labels. Why is this behavior desirable?

Answer:

Probabilities indicate confidence in predictions, which allows flexible decision-making using thresholds.

Scenario1: Penalizing big mistakes heavily

Question: A bank is predicting credit card spending limits. A few predictions are *very far* from the actual values, and the bank wants a metric that penalizes large errors more than small ones. Which metric should be used and why?

Answer: MSE (Mean Squared Error)

- Squares the errors → large errors grow exponentially
- Suitable when big mistakes are unacceptable
- Common in optimization because it's differentiable

Scenario2: Business wants “average mistake” in simple terms

Question: A sales team wants to know, on average, how much their demand prediction is off, in the same unit as sales. They don't want the metric to exaggerate outliers.

Answer: MAE (Mean Absolute Error)

- Errors are treated equally

- Easy to explain: *“We are wrong by $\sim X$ units on average”*
- Robust to outliers compared to MSE/RMSE

Scenario3: Occasional large errors matter, but not too extreme

Question: A ride-fare prediction model usually performs well but occasionally makes large mistakes during peak hours. The team wants a metric that penalizes large errors, but still stays interpretable.

Answer: RMSE (Root Mean Squared Error)

- Penalizes large errors (like MSE)
- Square root brings it back to original units
- Balanced choice between MAE and MSE

Scenario4: Comparing two models, not error size

Question: Two models predict house prices. One model’s error values look smaller, but management asks:

“How much better is this model compared to using just the average house price?”

Answer: R² Score (Coefficient of Determination)

- Measures **explained variance**
- Answers: *“How well does the model explain the data?”*
- Good for **model comparison**, not error magnitude

Scenario5: Dataset has many outliers

Question: A medical cost prediction dataset contains extreme values for a few patients. The model should not be dominated by these rare cases.

Answer: MAE

- Less sensitive to outliers
- Gives stable performance assessment.